Bipolar Junction Peltier Cooling in PIN Diodes

Why silicon is limited to millikelvin cooling, why germanium does better, and a roadmap towards a high-impact publication

Thesis project report

(COMSOL 6.3 Semiconductor + Heat Transfer study, reference model and analysis)

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Abstract

A forward-biased PIN diode absorbs heat where electrons and holes are injected over the junction barriers and releases heat where they recombine. With an engineered recombination zone at the centre of the intrinsic region, the junctions become net heat sinks. A 51-run COMSOL campaign and a COMSOL-independent ambipolar reference model show that in silicon the effect is real but tiny: 0.45 mK at 0.7 V and an estimated 20 mK–40 mK at the optimum bias near 0.9 V. This report explains why from first principles. (i) The heat absorbed per electron–hole pair is *independent of the band gap* and depends only on the carrier density. (ii) A "lever-arm" form of the heat equation shows that only recombination inside a small central fraction of each half of the device produces cooling. (iii) The achievable temperature drop scales as $n_\mathrm{b} D_\mathrm{a} \Pi_\mathrm{side}/k$, and Auger recombination caps the injected density. Together these give a closed-form ceiling $\Delta T_\mathrm{max} \propto (D_\mathrm{a}\Pi_\mathrm{side})^{3/2}/(k\,W\sqrt{E_\mathrm{pair} C_\mathrm{a}})$. Silicon's thermal conductivity ($k = 148\,\mathrm{W/(m\,K)}$) and modest ambipolar diffusivity ($D_\mathrm{a} \approx 18\,\mathrm{cm^2/s}$) put it in the millikelvin range. Germanium combines a 3.6× larger D_a with a 2.5× lower k. Its material figure of merit D_a/k is therefore 8.9× that of silicon, and the Auger-limited figure of merit is about 13–27× larger. The reference model predicts 0.15 K–0.40 K for a germanium device with the same geometry, at only 0.55 V and with a higher coefficient of performance. We close with five research directions grounded in the current literature, aimed at journals such as *Physical Review Applied*, *Nature Communications* and *Nature Electronics*, and with a phase-by-phase COMSOL roadmap.

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1 Introduction and what the simulations have shown so far

1.1 The device and the question

The device is a one-dimensional silicon PIN diode: a 2 µm n^+ layer and a 2 µm p^+ layer (both $1 \times 10^{19}/\mathrm{cm}^3$) separated by a $W_i = 1$ mm intrinsic region with cross-section $A_c = 1\,\mathrm{mm}^2$. Under forward bias, electrons are injected from the n^+ side and holes from the p^+ side. They flood the intrinsic region (high injection, $n \approx p \gg n_\mathrm{i}$) and recombine there. An optional recombination zone, a slab of short Shockley–Read–Hall (SRH) lifetime τ_short and width w_rz at the centre, decides *where* they recombine.

Two thermal configurations are used throughout:

- **Config A:** both ends held at T_room, with a sink at the centre at $T_\mathrm{room} + \Delta T_\mathrm{sink}$. It gives the heat Q_room pumped out of the room and ΔT_max, the sink temperature rise at which $Q_\mathrm{room} = 0$.
- **Config B:** the centre held at T_room and the ends thermally insulated. For the symmetric bar, $T_\mathrm{room} - T_\mathrm{end} = \Delta T_\mathrm{max}$ of Config A exactly.

The thermal problem is solved with one-way coupling: the semiconductor is solved at 300 K, and its heat source $Q = \boldsymbol{J}\cdot\boldsymbol{E} + U(E_\mathrm{g} + 3k_\mathrm{B}T)$ plus the contact Peltier terms drives the heat equation.

1.2 Summary of the results so far

Table 1 collects the key outputs.

Table 1: Selected COMSOL results of the 51-run matrix (0.7 V unless stated) and the extended-bias reference model. $\Delta T = T_\mathrm{room} - T_\mathrm{end}$ in Config B, or ΔT_max in Config A.

Run	Change	ΔT (mK)	Q_room (mW)	Note
V0	uniform lifetime, 1 ms	−0.152 (heats)	−0.090	room heated
V1	zone 100 µm/1 µs	0.447	0.263	COP = 0.74
V1	same, best over $V_\mathrm{F} \le 0.8$ V	2.38	–	at top of sweep
V3b	corrected heat source	0.445	0.262	$\eta_E = 1.000$
V4	mesh ×2	0.446	0.263	converged (0.19%)
V2b	FD + BGN + carrier–carrier scatt.	0.386	0.228	−14%
V6 best	zone 25 µm/1 µs	1.178	0.695	optimum at 0.7 V
V8b	Caughey–Thomas mobility, best	2.56	–	at 0.8 V
FINAL (ref. model)	zone 25 µm/0.3 µs, to 1.5 V	32.4	19.2	peak 0.92 V

Three facts stand out:

1. Uniform lifetime (V0) always heats the junction ends.
2. An engineered central zone turns the device into a cooler, but even after optimising zone and bias the cooling stays in the tens of millikelvin. The COMSOL/reference ratio of 0.35–0.63 seen in the matrix suggests a COMSOL value of about 10 mK–20 mK at the optimum.
3. Modelling choices (mesh, heat-source form, mobility model) change the answer by less than 15 %. The smallness is therefore physical, not numerical.

Sections 2–3 explain why. Section 4 shows why germanium is better, Section 5 proposes research directions, and Section 6 gives the COMSOL roadmap.

2 Physics of bipolar junction Peltier cooling

2.1 A qualitative picture first

Think of the junction barrier as a hill and the injected carriers as hikers.

- Only the most energetic electrons of the n^+ reservoir can climb over the barrier into the intrinsic region. When they leave, they take more than the average energy with them, and the lattice at the junction has to supply it. This is *evaporative cooling*, the same mechanism that cools a cup of coffee. The same happens to holes on the p^+ side.

- The injected electrons and holes then wander through the intrinsic region by *ambipolar diffusion*. Electrons and holes move together because any separation creates an electric field that pulls them back together. Eventually they recombine, and recombination releases the full energy of the pair, about one band gap plus their thermal energy, as heat.
- The junction is therefore a *cold source* and the recombination site a *hot source*. Whether the ends end up colder than the centre depends on two questions: *how far apart* the cold and hot sources are, and *how easily heat flows back* from the hot source to the cold one.

These two questions, geometry of recombination and thermal back-flow, govern everything that follows.

2.2 Energy book-keeping per electron–hole pair

For non-degenerate carriers the mean kinetic energy is $\frac{3}{2}k_{\mathrm{B}}T$ per carrier. A recombining pair therefore releases

$$E_{\mathrm{pair}} = E_{\mathrm{g}} + 3k_{\mathrm{B}}T \tag{1}$$

into the lattice. This is exactly the recombination term $U(E_{\mathrm{g}} + 3k_{\mathrm{B}}T)$ of COMSOL's heat source [3, 2].

The power supply delivers qV_{F} per pair. In the intrinsic region the electron and hole quasi-Fermi levels are split by qV_{j}, where V_{j} is the junction voltage: the part of V_{F} that drops across the two junctions, as opposed to the resistive drop V_i in the intrinsic region. Energy conservation per pair then requires the lattice on the *cold sides* (junctions and contacts together) to supply

$$\Pi_{\mathrm{tot}} \equiv E_{\mathrm{pair}} - qV_{\mathrm{j}}, \qquad \Pi_{\mathrm{side}} = \tfrac{1}{2}\Pi_{\mathrm{tot}} \ \text{ per side.} \tag{2}$$

This is the bipolar Peltier coefficient of the device [1]. Applying more voltage injects more pairs, but each pair absorbs *less* heat, because the barrier it climbs, $V_{\mathrm{bi}} - V_{\mathrm{j}}$, shrinks.

2.3 The heat absorbed per pair does not depend on the band gap

One might expect a wide-gap semiconductor to be a better cooler, since each pair carries E_{g} of energy. It is not, and the reason is instructive. To obtain a given carrier density n, a wide-gap material needs a correspondingly larger junction voltage. The gain in E_{pair} is cancelled exactly by the larger qV_{j}.

To show this, use the law of the junction in high injection. With $np = n_{\mathrm{i}}^2 \exp(qV_{\mathrm{j}}/k_{\mathrm{B}}T)$ and $n = p$,

$$n = n_{\mathrm{i}}\, e^{qV_{\mathrm{j}}/2k_{\mathrm{B}}T} \quad \Longrightarrow \quad qV_{\mathrm{j}} = 2k_{\mathrm{B}}T \ln \frac{n}{n_{\mathrm{i}}}. \tag{3}$$

Inserting the non-degenerate intrinsic density $n_{\mathrm{i}} = \sqrt{N_c N_v}\, e^{-E_{\mathrm{g}}/2k_{\mathrm{B}}T}$ into Eq. (2):

$$\boxed{\Pi_{\mathrm{tot}} = E_{\mathrm{g}} + 3k_{\mathrm{B}}T - 2k_{\mathrm{B}}T \ln \frac{n\, e^{E_{\mathrm{g}}/2k_{\mathrm{B}}T}}{\sqrt{N_c N_v}} = 3k_{\mathrm{B}}T + 2k_{\mathrm{B}}T \ln \frac{\sqrt{N_c N_v}}{n}} \tag{4}$$

Key result: band-gap independence

The band gap cancels. At a given carrier density, the heat absorbed per pair depends only on temperature and on the effective densities of states. At $n = 1 \times 10^{17}/\mathrm{cm}^3$ and 300 K:

- **Si** ($\sqrt{N_c N_v} = 1.71 \times 10^{19}/\mathrm{cm}^3$): $\Pi_{\mathrm{tot}} = 0.343\,\mathrm{eV}$.
- **Ge** ($\sqrt{N_c N_v} = 7.9 \times 10^{18}/\mathrm{cm}^3$): $\Pi_{\mathrm{tot}} = 0.304\,\mathrm{eV}$.

The heat *per pair* is therefore not where materials differ. They differ in how many pairs can be pushed across per second (transport), how fast heat flows back (thermal conductivity), and how much voltage this costs (efficiency).

2.4 The lever-arm equation: where recombination happens matters

Intuition. Consider one half of the bar in Config B. The end at $x = 0$ is the cold junction, and the centre at $x = L_\mathrm{h} = W_i/2$ is held at T_room. Heat released at a point x must flow to the centre sink. On its way it raises the temperature of everything between x and the centre, but *not* the region beyond x towards the end. Seen from the cold end, a heat source at x therefore acts through a "lever arm" $(L_\mathrm{h} - x)$. Heat released near the end counts fully; heat released at the centre does not count at all.

Derivation. In steady state, Fourier's law $q_x = -k\,\mathrm{d}T/\mathrm{d}x$ and energy conservation $\mathrm{d}q_x/\mathrm{d}x = H(x)$ hold, with H the volumetric heat source. The cold end absorbs the Peltier flux s, so $q_x(0) = -s$, and $T(L_\mathrm{h}) = T_\mathrm{room}$. Integrating twice gives

$$\boxed{\Delta T \equiv T_\mathrm{room} - T(0) = \frac{1}{k}\left[\, s\,L_\mathrm{h} - \int_0^{L_\mathrm{h}} (L_\mathrm{h} - x)\,H(x)\,\mathrm{d}x \right]} \tag{5}$$

In the symmetric device, every pair injected through this end recombines in this half. The pair flux $\Gamma = J/q$ is therefore $\Gamma = \int_0^{L_\mathrm{h}} R(x)\,\mathrm{d}x$, where R is the recombination rate. The Peltier flux is $s = \Pi_\mathrm{side}\Gamma$, and the heat source is $H = E_\mathrm{pair}R + \boldsymbol{J}\cdot\boldsymbol{E}$. Substituting,

$$\Delta T = \frac{1}{k}\int_0^{L_\mathrm{h}} R(x)\Big[\underbrace{\Pi_\mathrm{side}L_\mathrm{h}}_{\text{cooling credit}} - \underbrace{E_\mathrm{pair}(L_\mathrm{h} - x)}_{\text{heating debit}}\Big]\mathrm{d}x \;-\; \Delta T_\mathrm{J}, \tag{6}$$

where ΔT_J is the Joule contribution. Each recombination event earns a fixed cooling credit and pays a heating debit that grows with its distance from the centre. It produces net cooling only if it happens beyond the break-even point

$$x^* = L_\mathrm{h}\left(1 - \frac{\Pi_\mathrm{side}}{E_\mathrm{pair}}\right). \tag{7}$$

Physical insight: why uniform lifetime can never cool

With $\Pi_\mathrm{side} = \frac{1}{2}(E_\mathrm{pair} - qV_\mathrm{j})$, we have $\Pi_\mathrm{side}/E_\mathrm{pair} < \frac{1}{2}$ always. For silicon at the optimum ($V_\mathrm{j} \approx 0.85\,\mathrm{V}$), $\Pi_\mathrm{side}/E_\mathrm{pair} \approx 0.14$, so only recombination in the **innermost** 14 % of each half cools. If recombination were uniform, the average lever arm would be $L_\mathrm{h}/2$, and the bracket in Eq. (6) would average to $L_\mathrm{h}(\Pi_\mathrm{side} - E_\mathrm{pair}/2) < 0$. *Uniform recombination always heats the ends.* In a real uniform-lifetime diode the carrier density, and with it the recombination, is highest near the junctions, which makes it worse. This is exactly the V0 result, and it is why the recombination zone is indispensable.

2.5 Diffusion-limited pumping: the scaling $\Delta T \simeq n_\mathrm{b}D_\mathrm{a}\Pi_\mathrm{side}/k$

Intuition. Suppose the zone were a perfect sink at the centre. All the cooling credit then comes from how many pairs reach the centre per second. In high injection, pairs move by ambipolar diffusion down a concentration gradient, so the pair flux is set by the boundary density n_b and the diffusivity. Back-conduction of heat is set by k.

Equations. For a perfect central sink the profile is linear, $n(x) = n_\mathrm{b}(1 - x/L_\mathrm{h})$, and

$$\Gamma = \frac{D_\mathrm{a}n_\mathrm{b}}{L_\mathrm{h}}, \qquad D_\mathrm{a} = \frac{2D_nD_p}{D_n + D_p} = \frac{k_\mathrm{B}T}{q}\,\frac{2\mu_n\mu_p}{\mu_n + \mu_p}. \tag{8}$$

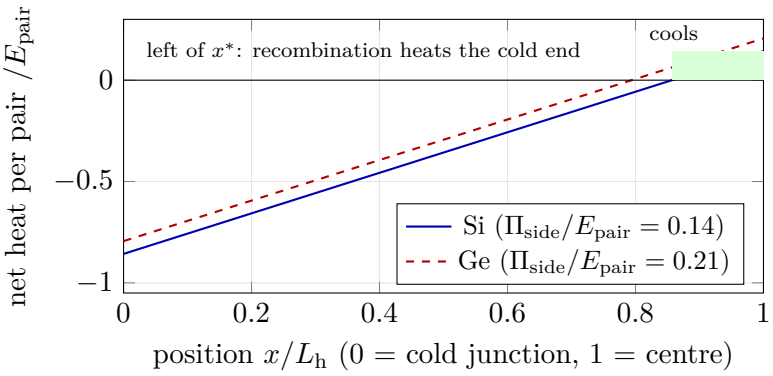


Figure 1: Lever-arm weight $[\Pi_{\mathrm{side}} L_{\mathrm{h}} - E_{\mathrm{pair}}(L_{\mathrm{h}} - x)]/(E_{\mathrm{pair}} L_{\mathrm{h}})$ of Eq. (6), evaluated at each material's optimum bias ($n \approx 1 \times 10^{17}/\mathrm{cm}^3$). Recombination contributes net cooling only to the right of x^*.

Putting all recombination at $x = L_{\mathrm{h}}$ in Eq. (6) gives

$$\boxed{\Delta T_{\mathrm{ideal}} = \frac{L_{\mathrm{h}}\,\Gamma\,\Pi_{\mathrm{side}}}{k} = \frac{n_{\mathrm{b}}\,D_{\mathrm{a}}\,\Pi_{\mathrm{side}}}{k}} \tag{9}$$

The device length cancels: a shorter device pumps more heat ($\Gamma \propto 1/L_{\mathrm{h}}$) but also conducts more heat back, by the same factor. The V5 runs at 0.7 V confirm this length independence.

A zone of finite strength, with sink velocity $S \approx w_{\mathrm{rz}}/(2\tau_{\mathrm{short}})$ per side, acts as a series "resistance" to the pair flux:

$$\Gamma = \frac{n_{\mathrm{b}}}{L_{\mathrm{h}}/D_{\mathrm{a}} + 1/S} \quad\Longrightarrow\quad \Delta T = \frac{n_{\mathrm{b}} D_{\mathrm{a}} \Pi_{\mathrm{side}}}{k}\,\frac{1}{1 + D_{\mathrm{a}}/(S L_{\mathrm{h}})}. \tag{10}$$

Cooling therefore needs $S \gg D_{\mathrm{a}}/L_{\mathrm{h}}$, which is 360 cm/s for silicon. This is consistent with the V6 optimum, $S \approx 500\,\mathrm{cm/s} - -2500\,\mathrm{cm/s}$. If the zone is too strong, it drains the carriers, the intrinsic region becomes resistive, and at fixed V_{F} the junction voltage, and with it n_{b}, falls.

2.6 The Auger ceiling

Intuition. Raising the bias raises n_{b} exponentially. Why, then, is there an optimum? Auger recombination, in which three carriers are involved, grows as n^3. It happens wherever the density is high, which is right next to the cold junction, where the lever arm is longest. Beyond a certain density these "wrong-place" recombination events dominate.

Equations. Take the Auger rate $R_A = C_{\mathrm{a}} n^3$ (ambipolar coefficient C_{a}) on the linear profile and insert it as a heating debit into Eq. (5):

$$\Delta T_A = \frac{E_{\mathrm{pair}} C_{\mathrm{a}}}{k} \int_0^{L_{\mathrm{h}}} n_{\mathrm{b}}^3 \Big(1 - \frac{x}{L_{\mathrm{h}}}\Big)^3 (L_{\mathrm{h}} - x)\,\mathrm{d}x = \frac{E_{\mathrm{pair}} C_{\mathrm{a}} n_{\mathrm{b}}^3 L_{\mathrm{h}}^2}{5k}. \tag{11}$$

The net temperature drop as a function of the injected density is then

$$\Delta T(n_{\mathrm{b}}) \simeq \frac{1}{k}\left[\Pi_{\mathrm{side}} D_{\mathrm{a}} n_{\mathrm{b}} - \frac{E_{\mathrm{pair}} C_{\mathrm{a}} L_{\mathrm{h}}^2}{5} n_{\mathrm{b}}^3\right]. \tag{12}$$

It is maximal at

$$n_{\mathrm{opt}} = \frac{1}{L_{\mathrm{h}}}\sqrt{\frac{5\,\Pi_{\mathrm{side}} D_{\mathrm{a}}}{3\,E_{\mathrm{pair}} C_{\mathrm{a}}}}, \qquad \boxed{\Delta T_{\mathrm{max}}^{\mathrm{Auger}} = \frac{2}{3}\frac{n_{\mathrm{opt}} D_{\mathrm{a}} \Pi_{\mathrm{side}}}{k} = \frac{2}{3\,k\,L_{\mathrm{h}}}\,(\Pi_{\mathrm{side}} D_{\mathrm{a}})^{3/2}\sqrt{\frac{5}{3\,E_{\mathrm{pair}} C_{\mathrm{a}}}}} \tag{13}$$

Two consequences follow. First, the material enters through the combination

$$\mathcal{F} \equiv \frac{(D_{\mathrm{a}}\Pi_{\mathrm{side}})^{3/2}}{k\sqrt{E_{\mathrm{pair}}C_{\mathrm{a}}}}, \tag{14}$$

which we call the *Auger-limited figure of merit*. Second, at the Auger limit the length *does* matter: $\Delta T_{\max} \propto 1/W_i$, because a shorter device needs less density for the same flux. This is not seen at 0.7 V, where the device is far from the Auger limit.

2.7 Joule heating is a minor term

With $\sigma(x) = q(\mu_n + \mu_p)n(x)$ on the linear profile, the Joule contribution to Eq. (5) is

$$\Delta T_{\mathrm{J}} = \frac{1}{k}\int_0^{L_{\mathrm{h}}} \frac{J^2}{\sigma(x)}(L_{\mathrm{h}} - x)\,\mathrm{d}x = \frac{J^2 L_{\mathrm{h}}^2}{q(\mu_n + \mu_p)n_{\mathrm{b}}k}. \tag{15}$$

Relative to Eq. (9), using Eq. (8),

$$\frac{\Delta T_{\mathrm{J}}}{\Delta T_{\mathrm{ideal}}} = \frac{k_{\mathrm{B}}T}{\Pi_{\mathrm{side}}}\,\frac{2\mu_n\mu_p}{(\mu_n + \mu_p)^2} \approx \begin{cases} 0.06 & \text{Si} \\ 0.08 & \text{Ge} \end{cases} \tag{16}$$

Joule heating in the intrinsic region costs only a few per cent. The limits are the lever arm, the Auger rate, and the ratio D_{a}/k. Joule heating takes over only beyond the optimum, when $V_{\mathrm{j}} \to V_{\mathrm{bi}}$ and every extra volt drops resistively; this is the heating seen above about 1 V in the extended sweep.

3 Why silicon is limited to millikelvin cooling

3.1 Numbers

For silicon the parameters are
$D_{\mathrm{a}} = 18.3\,\mathrm{cm}^2/\mathrm{s}$ ($\mu_n = 1414$, $\mu_p = 471\,\mathrm{cm}^2/(\mathrm{V\,s})$), $k = 148\,\mathrm{W/(m\,K)}$ [5], $E_{\mathrm{pair}} = 1.198\,\mathrm{eV}$, $\Pi_{\mathrm{side}} \approx 0.172\,\mathrm{eV}$ at $1 \times 10^{17}/\mathrm{cm}^3$, and the Dziewior–Schmid $C_{\mathrm{a}} = C_n + C_p = 3.8 \times 10^{-31}\,\mathrm{cm}^6/\mathrm{s}$ [6]. Equation (13) then gives

$$n_{\mathrm{opt}} \approx 6.8 \times 10^{16}/\mathrm{cm}^3, \qquad \Delta T_{\max}^{\mathrm{Auger}} \approx 15\,\mathrm{mK}. \tag{17}$$

The ambipolar Auger coefficient measured at high injection, $C_{\mathrm{a}} = 1.66 \times 10^{-30}\,\mathrm{cm}^6/\mathrm{s}$ [7], gives 7 mK. The full reference model, which also credits the Peltier heat of Auger pairs that cross the junction, gives 32 mK–37 mK (Dziewior–Schmid) or 21 mK (high-injection Auger). The analytic ceiling is within a factor of about 2.4 of the full model, and both say "tens of millikelvin at best".

3.2 The reason in one sentence

Key result: why silicon gives millikelvin

Silicon conducts heat about a hundred times too well for the rate at which ambipolar diffusion can deliver pairs to the recombination zone, and Auger recombination stops us from compensating with a higher carrier density. With $D_{\mathrm{a}}/k = 1.2 \times 10^{-5}\,\mathrm{m}^3\,\mathrm{K/J}$, even $n_{\mathrm{b}} = 1 \times 10^{17}/\mathrm{cm}^3$ and $\Pi_{\mathrm{side}} = 0.17\,\mathrm{eV}$ give only $n_{\mathrm{b}}D_{\mathrm{a}}\Pi_{\mathrm{side}}/k \approx 34\,\mathrm{mK}$ in the ideal limit.

3.3 An equivalent view: the device as a poor thermoelectric

A thermoelectric leg reaches $\Delta T_{\max} = \frac{1}{2} Z T_c^2$ with $Z = S^2\sigma/k$ [11]. Define for the PIN an effective Seebeck coefficient $S_{\text{eff}} = \Pi_{\text{side}}/(qT)$ and a conductivity $\sigma = q n_{\text{b}}(\mu_n + \mu_p)$. At $n_{\text{b}} = 1 \times 10^{17}/\text{cm}^3$:

	S_{eff} (µV/K)	σ (S/m)	ZT	$\frac{1}{2}ZT^2$ (K)
Si PIN	573	3.0e3	0.002	0.30
Ge PIN	507	9.3e3	0.012	1.8
Bi_2Te_3 (for scale)	∼200	∼1e5	∼1	∼70

The Seebeck coefficient of the bipolar junction is *large*, about 3× that of Bi_2Te_3. What is fatal is σ/k, which is about 1000 times worse. The real device stays about 10× below even this optimistic bound, because transport is diffusive and recombining, not ohmic. Silicon's $ZT \approx 0.002$ is the thermoelectric way of saying "millikelvin".

3.4 Can engineering rescue silicon?

- **Shorter devices:** $\Delta T_{\max}^{\text{Auger}} \propto 1/W_i$, but the current density scales as $\Gamma \propto n_{\text{opt}}/L_{\text{h}} \propto 1/W_i^2$. A 20× gain ($W_i = 50\,\mu\text{m}$) would require current densities of order $10^4\,\text{A/cm}^2$ and $\text{kW}\,\text{cm}^{-2}$ heat fluxes. This is not practical.
- **Higher bias:** beyond the optimum ($\approx 0.92\,\text{V}$) Auger heat and the vanishing barrier ($V_{\text{j}} \to V_{\text{bi}}$) turn cooling into heating (Fig. 2).
- **Lower k by nanostructuring** (holey Si $\sim 2\,\text{W}/(\text{m}\,\text{K})$ [13], rough nanowires [12]): $\Delta T \propto 1/k$, so this is the only strong lever. The surfaces it introduces, however, destroy the lifetimes and mobilities on which D_{a} and the zone design rely, unless they are passivated to $S \lesssim 10\,\text{cm/s}$.

4 Why germanium gives better results

4.1 Intuition

Equation (4) showed that the band gap is irrelevant for the heat per pair, and Eqs. (9) and (13) showed what matters: carriers that diffuse fast (D_{a}), a lattice that conducts heat poorly (k), and weak Auger recombination (C_{a}). Germanium is a better conductor of *carriers*, with both electron and hole mobilities 2.8–4 times higher than in silicon, and a worse conductor of *heat*, because of its heavier atoms and softer lattice. Its narrow gap gives an additional, independent bonus: by Eq. (3), the same carrier density is reached at roughly half the voltage, so each pair costs half the electrical energy.

4.2 Material parameters

4.3 Mathematical comparison

Diffusion-limited regime. At equal n_{b}, Eq. (9) gives

$$\frac{\Delta T_{\text{Ge}}}{\Delta T_{\text{Si}}} = \frac{(D_{\text{a}}/k)_{\text{Ge}}}{(D_{\text{a}}/k)_{\text{Si}}} \cdot \frac{\Pi_{\text{side}}^{\text{Ge}}}{\Pi_{\text{side}}^{\text{Si}}} = 8.9 \times 0.88 \approx 7.8. \tag{18}$$

Auger-limited optimum. From Eq. (13),

$$\frac{\Delta T_{\max}^{\text{Ge}}}{\Delta T_{\max}^{\text{Si}}} = \left[\frac{(D_{\text{a}}\Pi_{\text{side}})_{\text{Ge}}}{(D_{\text{a}}\Pi_{\text{side}})_{\text{Si}}}\right]^{3/2} \frac{k_{\text{Si}}}{k_{\text{Ge}}} \sqrt{\frac{(E_{\text{pair}}C_{\text{a}})_{\text{Si}}}{(E_{\text{pair}}C_{\text{a}})_{\text{Ge}}}} = 3.2^{3/2} \times 2.47 \times \begin{cases} 0.94 \\ 1.96 \end{cases} \tag{19}$$

which is about 13 when silicon is given the Dziewior–Schmid Auger coefficient, and about 27 when it is given the high-injection value. In absolute terms the formula gives $n_{\text{opt}} \approx 1.1 \times 10^{17}/\text{cm}^3$ and $\Delta T_{\max}^{\text{Auger}} \approx 0.20\,\text{K}$ for germanium.

Table 2: Parameters at 300 K used in this report.

Quantity	Si	Ge	Source
E_g (eV)	1.12	0.66	[4]
N_c, N_v ($1/\mathrm{cm}^3$)	2.8e19, 1.04e19	1.04e19, 6.0e18	COMSOL library [3], [4]
n_i ($1/\mathrm{cm}^3$)	9.65e9 (ref. model)	2.0e13	[4]
μ_n, μ_p ($\mathrm{cm}^2/(\mathrm{V\,s})$)	1414, 471	3900, 1900	[9, 4]
D_a (cm^2/s)	18.3	66.0	Eq. (8)
k ($\mathrm{W}/(\mathrm{m\,K})$)	148	60	[5]
C_a (cm^6/s)	3.8e-31 / 1.66e-30	7e-31	[6, 7], [8]
$E_{\mathrm{pair}} = E_g + 3k_BT$ (eV)	1.198	0.738	Eq. (1)
Π_{tot} at $1\times10^{17}/\mathrm{cm}^3$ (eV)	0.343	0.304	Eq. (4)
V_j for $1\times10^{17}/\mathrm{cm}^3$ (V)	0.85	0.44	Eq. (3)
$M = D_a/k$ ($\mathrm{m}^3\,\mathrm{K/J}$)	1.24e-5	1.10e-4	ratio **8.9**
$\mathcal{F}$, Eq. (14) (relative)	1	**13** / **27**	vs. Si C_a (DS / high inj.)

Voltage and efficiency. Per pair, the electrical cost is $qV_F \gtrsim qV_j$ and the cold-side gain is Π_{tot}. The ideal per-pair coefficient of performance, ignoring back-conduction, is

$$\mathrm{COP}_0 \simeq \frac{\Pi_{\mathrm{tot}}}{qV_F} \approx \frac{E_{\mathrm{pair}} - qV_j}{qV_j} \approx \begin{cases} 0.40 & \text{Si} \\ 0.68 & \text{Ge} \end{cases} \qquad (n = 1\times10^{17}/\mathrm{cm}^3). \tag{20}$$

Germanium therefore cools with about 1.7 times less electrical work per unit of heat.

4.4 Reference-model results

The COMSOL-independent reference model, which solves the ambipolar boundary-value problem with SRH, Auger and the exact heat integration, was run for both materials with identical geometry ($W_i = 1\,\mathrm{mm}$, 25 µm zone, $\tau_{\mathrm{long}} = 1\,\mathrm{ms}$). Germanium was given the parameters of Table 2.

Table 3: Peak cooling over bias (Config B, reference model).

Case	τ_{short}	peak ΔT	at V_F	COP	P_{el}
Si, C_a (DS)	0.3 µs	32 mK	0.92 V	0.20	96 mW
Si, C_a (DS)	0.1 µs	37 mK	0.97 V	0.20	110 mW
Si, C_a (high injection)	0.1 µs	22 mK	0.94 V	0.23	55 mW
Ge	0.1 µs	348 mK	0.55 V	0.28	295 mW
Ge	0.03 µs	397 mK	0.61 V	0.26	365 mW
Ge, mobilities ×0.5 (carrier–carrier)	0.1 µs	149 mK	0.57 V	0.29	125 mW
Ge, $W_i = 0.5\,\mathrm{mm}$	0.01 µs	600 mK	0.69 V	0.20	1.48 W

Key result: germanium

At identical geometry and an optimised zone, germanium cools about 10× more than silicon (0.35 K–0.40 K against 0.03 K–0.04 K in the reference model). It does so at about 0.55 V instead of 0.92 V, with a higher COP, and in agreement with the analytic ratio of 8–13. Including the strong carrier–carrier scattering recently measured in Ge at high injection [8] (about ×0.5 on the mobilities) gives 0.15 K. The realistic expectation, after COMSOL's typical factor of 0.4–0.6 with respect to the reference model, is therefore **about 0.1–0.2 K for Ge, against 0.01–0.02 K for Si**: one order of magnitude, but still sub-kelvin.

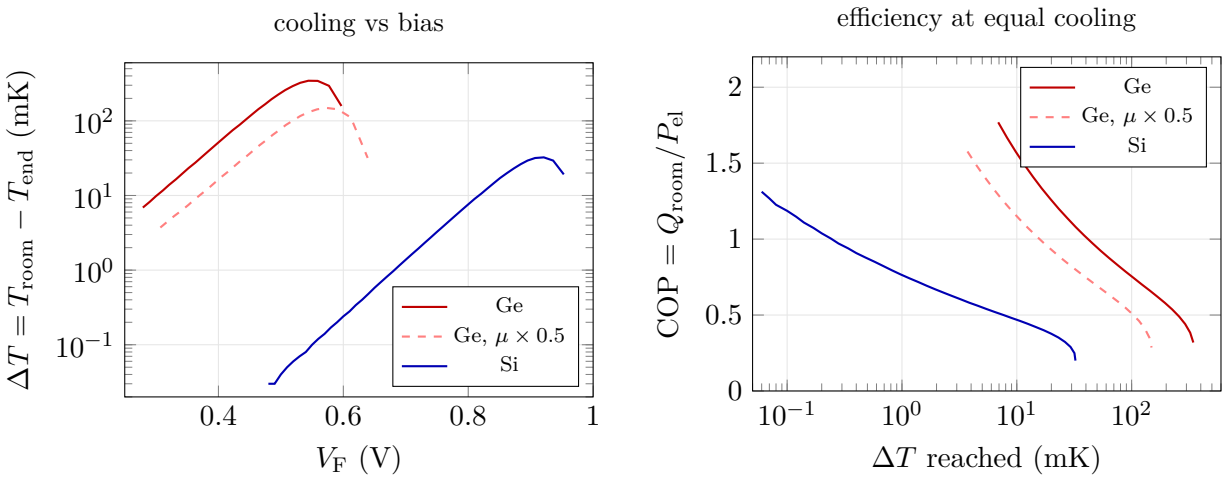


Figure 2: Reference-model comparison at identical geometry. Si: 25 µm/0.3 µs zone (the FINAL design). Ge: 25 µm/0.1 µs. Left: germanium cools about 10× more, at about 0.4 V lower bias. Right: COP against the cooling reached (rising branch up to each peak); at any given cooling level germanium needs less electrical power. Only positive ΔT is shown on the log scale; both devices heat beyond their peak.

4.5 What could go wrong with germanium

Caveat

1. **Carrier–carrier scattering.** The sum $\mu_n + \mu_p$ falls roughly as $8 \times 10^{20}/\Delta n\,\text{cm}^2/(\text{V s})$ at high injection [8], which bites exactly near n_{opt}. It must be modelled (Fletcher model [10] or a user-defined mobility).
2. **Lifetime.** The bulk lifetime must exceed the transit time $L_{\text{h}}^2/D_{\text{a}} \approx 40\,\mu\text{s}$ by a wide margin. Bulk high-purity Ge can provide this; Ge-on-Si epitaxy, with its threading dislocations, cannot.
3. **Contacts.** With $N_v = 6 \times 10^{18}/\text{cm}^3$, a p^+ layer at $1 \times 10^{19}/\text{cm}^3$ is degenerate. Fermi–Dirac statistics, or a lower doping (about $3 \times 10^{18}/\text{cm}^3$), and recomputed contact Peltier terms are needed. Metal/n-Ge contacts suffer from strong Fermi-level pinning [16].
4. **Thermal feedback.** $P_{\text{el}} \approx 0.3\,\text{W/mm}^2$ and $\Delta T \sim 0.3\,\text{K}$ approach the limit of one-way coupling. Since $\text{d}\ln n_{\text{i}}/\text{d}T \approx 4.8\,\%/\text{K}$, two-way electro-thermal coupling is required.
5. **Plausibility threshold.** The project's automatic check flags cooling of 0.1 K or more as a bug; that threshold was set for silicon. For germanium it must be raised to about 1 K.

5 Novel research directions for a high-impact publication

The minority-carrier (bipolar) Peltier effect has a solid theoretical basis [1, 2], and recent work keeps returning to the role of recombination in p–n thermoelectricity [20] and to junction-based thermoelectric devices [21]. Two gaps stand out. First, *direct experimental verification remains limited* [1]. Second, no work we are aware of gives *design rules* that connect recombination engineering, Auger limits and material choice to the achievable temperature drop. Several of the results in Sections 2–4 are new in this compact form. The following directions build on them.

5.1 N1 — A design theory of bipolar Peltier cooling (theory + simulation)

Idea. Publish the lever-arm framework: Eqs. (4), (6), (7), (10) and (13), the theorem that uniform recombination cannot cool, the band-gap independence of Π_{tot}, and the figures of merit M and $\mathcal{F}$. Validate the framework with the COMSOL matrix and the reference model, then

go one step further. Treat the lifetime profile $\tau(x)$ as a free function and find the profile that maximises ΔT (inverse design). Equation (6) makes ΔT a linear functional of $R(x)$, so the optimum can be computed by adjoint or gradient methods.

Why novel. Existing treatments compute the effect for given devices [1]; none provides closed-form ceilings or an optimal-profile theory.

Target. *Physical Review Applied*, *Journal of Applied Physics*, *IEEE Transactions on Electron Devices*.

5.2 N2 — First direct observation of bipolar Peltier cooling in a germanium PIN

Idea. Germanium brings the effect from about 10 mK to about 0.1 K. That is well within the reach of thermoreflectance imaging, which already separates the Peltier and Joule components of microcooler temperature maps with submicron resolution [18] (lock-in implementations typically resolve about 10 mK), and of Raman thermometry. Fabricate a lateral Ge PIN on high-purity Ge, form the central recombination zone by masked light-ion (H^+/He^{2+}) irradiation, as is standard for local lifetime control in power diodes [17], and image the end-to-centre temperature profile against bias. Compare devices with and without the zone.

Why novel. It would be a clean demonstration of minority-carrier thermoelectric cooling controlled by recombination engineering, a long-standing prediction [1]. The zone/no-zone sign reversal (V1 vs. V0) is an unambiguous fingerprint that Joule heating cannot mimic.

Target. *Nature Communications*, *Nature Electronics*; *Physical Review Applied* as a fallback. A high-impact venue requires the experiment; simulation alone is not enough.

5.3 N3 — "Carrier crystal, phonon glass": a materials map towards kelvin cooling

Idea. Since $\Delta T \propto D_\mathrm{a}/k$, the target is a semiconductor that keeps germanium-like carrier diffusion but scatters phonons strongly. GeSn alloys are a natural candidate: their lattice thermal conductivity drops from about 55 to about 4 W/(m K) at 12–14% Sn [14, 15]. In the reference model, a Ge-like device with $k = 4\,\mathrm{W/(m\,K)}$ reaches about 5 K with full Ge mobilities, and about 2 K with mobilities reduced to $D_\mathrm{a} = 30\,\mathrm{cm^2/s}$. This is the first route to *kelvin-scale* bipolar cooling that we can identify. The publishable product is a design map of iso-ΔT contours over $(k, D_\mathrm{a}, C_\mathrm{a}, \tau)$, with Si, Ge, SiGe and GeSn placed on it, and with the lifetime and mobility each alloy must retain stated explicitly. The criterion follows from Eq. (9):

$$\Delta T \gtrsim 1\,\mathrm{K} \iff \frac{D_\mathrm{a}}{k} \gtrsim \frac{1\,\mathrm{K}}{\eta\, n_\mathrm{b} \Pi_\mathrm{side}} \approx 8 \times 10^{-4}\,\mathrm{m^3\,K/J}, \tag{21}$$

for $n_\mathrm{b} = 1 \times 10^{17}/\mathrm{cm^3}$, $\Pi_\mathrm{side} = 0.15\,\mathrm{eV}$ and a model-to-ideal efficiency $\eta \approx 0.5$.

Risk. Epitaxial GeSn currently has short lifetimes and defect-limited mobility. The map makes that bottleneck quantitative, which is valuable in itself.

Target. *Physical Review Applied*, *Advanced Functional Materials*, *ACS Applied Energy Materials*.

5.4 N4 — Transient (pulsed) bipolar supercooling

Idea. Peltier absorption at the junction is immediate, whereas the recombination heat released in the zone needs the thermal diffusion time $t_\mathrm{th} = L_\mathrm{h}^2/\alpha$ to reach the end. This is 2.8 ms for Si and 7 ms for Ge ($\alpha_\mathrm{Ge} = 3.5 \times 10^{-5}\,\mathrm{m^2/s}$). Pair transit to the zone is faster: $t_c \approx L_\mathrm{h}^2/2D_\mathrm{a} \approx 20\,\mu\mathrm{s}$ in Ge. A current pulse that overdrives the steady-state optimum for $t_c \ll t \ll t_\mathrm{th}$ should therefore produce a transient cooling that exceeds the steady state. This is analogous to the supercooling

of thermoelectric coolers by current pulses [19]. For a constant absorbed flux s on a semi-infinite solid, the surface cools as

$$\Delta T_s(t) = \frac{2s}{e}\sqrt{\frac{t}{\pi}}, \qquad e = \sqrt{k\rho c_p}, \tag{22}$$

with $e_{\mathrm{Ge}} \approx 1.0 \times 10^4$ against $e_{\mathrm{Si}} \approx 1.6 \times 10^4\,\mathrm{W\,s^{1/2}/(m^2\,K)}$, which again favours Ge.

Target. *Applied Physics Letters*, *Physical Review Applied*.

5.5 N5 — Cooling below ambient in a self-contained device by exporting energy as light

Idea. Config C (lateral sink, no external cold reservoir) showed that the ends cannot fall below ambient. The whole electrical power must leave through the same sink, so the bar warms by about $P_{\mathrm{el}}/(h_{\mathrm{lat}}PL)$, which exceeds the local Peltier dip. If the recombination zone is made radiative, for example with direct-gap GeSn ($\gtrsim$6–8% Sn [25]) or a III–V zone, part of the pair energy leaves as photons. The condition for sub-ambient ends becomes

$$|\Delta T_{\mathrm{dip}}| > \frac{P_{\mathrm{el}} - P_{\mathrm{rad}}}{h_{\mathrm{lat}}\,P\,L_{\mathrm{sink}}}, \tag{23}$$

which links junction Peltier cooling to electroluminescent and thermophotonic cooling [22, 23]. This is speculative: the external quantum efficiency of Ge-based emitters is low, and silicon is excluded [24]. It is a bold concept for a theory paper.

5.6 Summary of directions

	Contribution	Evidence needed	Realistic venue
N1	Lever-arm theory, ceilings, inverse-designed $\tau(x)$	COMSOL + reference model	PR Applied / JAP
N2	First direct measurement, Ge PIN with irradiated zone	Fabrication + thermoreflectance	Nat. Commun. / Nat. Electron.
N3	Materials map towards kelvin cooling (GeSn)	Simulation + literature data	PR Applied / AFM
N4	Pulsed supercooling	Transient COMSOL	APL / PR Applied
N5	Radiative export for sub-ambient cooling	Coupled optical–electrical–thermal model	PR Applied (theory)

Suggested sequence: N1 together with the Ge part of N3 as the first paper (fully within the current COMSOL framework), then N4. N2 should be pursued with an experimental collaborator.

6 Roadmap for the COMSOL simulations

Every phase reuses the verified framework (Driver A, the configuration files, the post-processing checks) and ends with explicit pass/fail criteria.

Phase 0 — Close the silicon chapter (1 week)

- Run `run_final.py` (FINAL, V1x, optional zone scan) up to 1.5 V, then `final_report.py`.

- *Checks:* $\eta_E \in [0.98, 1.02]$ (corrected heat source); peak $\Delta T < 0.1\,\mathrm{K}$; COMSOL/reference ratio between 0.3 and 1; self-heating flag below about 1.2 V.
- *Deliverable:* the silicon numbers for Table 1 and Fig. 2, with COMSOL replacing the reference model.

Phase 1 — Germanium device, one-way coupling (2 weeks)

- **Material:** Ge from the COMSOL Semiconductor Material Library, or user-defined properties: $\varepsilon_r = 16.2$, $E_\mathrm{g} = 0.66\,\mathrm{eV}$, $\chi = 4.0\,\mathrm{eV}$, $N_c = 1.04 \times 10^{19}$, $N_v = 6.0 \times 10^{18}/\mathrm{cm}^3$, $\mu_n = 3900$, $\mu_p = 1900\,\mathrm{cm}^2/(\mathrm{V\,s})$, $k = 60\,\mathrm{W/(m\,K)}$, $\rho = 5323\,\mathrm{kg/m}^3$, $c_p = 320\,\mathrm{J/(kg\,K)}$.
- **Recombination:** Auger $C_n = C_p = 3.5 \times 10^{-31}\,\mathrm{cm}^6/\mathrm{s}$ (ambipolar 7×10^{-31} [8]); SRH $\tau_\mathrm{long} = 1\,\mathrm{ms}$.
- **Doping and contacts:** contacts at $3 \times 10^{18}/\mathrm{cm}^3$, so that Boltzmann statistics remain valid. Recompute $\Pi_{nc}, \Pi_{pc}, \Pi_{rn}, \Pi_{rp}$ with the Ge values of N_c, N_v and E_g.
- **Mesh:** the extrinsic Debye length is about 3 nm at $3 \times 10^{18}/\mathrm{cm}^3$, so keep about 1 nm elements at the junctions and mesh factor 2 in the zone.
- **Sweeps:** V_F from 0 to 0.9 V in steps of 0.005 V; ΔT_sink in $\pm 1\,\mathrm{K}$.
- **Zone scan:** $w_\mathrm{rz} \in \{25, 50\}\,\mu\mathrm{m} \times \tau_\mathrm{short} \in \{0.01, 0.03, 0.1, 0.3\}\,\mu\mathrm{s}$, and $W_i \in \{0.25, 0.5, 1\}\,\mathrm{mm}$.
- *Checks:* energy balance, mesh convergence (less than 2%), Ge reference-model ratio, plausibility threshold raised to 1 K.
- *Deliverable:* the COMSOL Si-vs-Ge comparison (core of paper N1).

Phase 2 — Realistic transport (2 weeks)

- Carrier–carrier scattering: the Fletcher model [10] with Ge coefficients, calibrated against the high-injection mobility of Ref. [8].
- Fermi–Dirac statistics and band-gap narrowing, with contact Peltier terms computed from the *local* band edges at the contacts, $\Pi_n = (E_c - E_{Fn})/q + 2k_\mathrm{B}T/q$, so that η_E stays at 1.
- Report each effect separately, as was done for silicon in V2 and V8.

Phase 3 — Two-way electro-thermal coupling (1–2 weeks)

- Feed the Heat Transfer temperature into the semiconductor model inputs.
- Add $E_\mathrm{g}(T)$ (Varshni [26]), $\mu(T)$ and $k(T)$ (Ge: k falls roughly as $1/T$ near room temperature [5]).
- Use a fully coupled solver with V_F continuation. Compare with one-way coupling. Required for Ge, because ΔT and the self-heating approach a kelvin.

Phase 4 — Transient pulsed operation, N4 (2 weeks)

- Time-dependent study with a smoothed rectangular current pulse (amplitude 1–10× the steady optimum, width 10 µs–10 000 µs).
- Time steps well below t_c at the edges of the pulse.
- Outputs: $T_\mathrm{end}(t)$, the minimum transient temperature against pulse parameters, and a comparison with the steady state.

Phase 5 — A realistic 2D device for experiment, N2 (3 weeks)

- Lateral Ge PIN on a Ge-on-insulator (GOI) or suspended Ge membrane, 2 µm–10 µm thick.
- Metal contacts with specific contact resistance, passivated surfaces (surface recombination velocity 10 cm/s–100 cm/s), and an implanted zone defined by a mask.
- Heat sink through the substrate under the zone; surface temperature map for comparison with thermoreflectance.
- Boundary-layer meshing at the surfaces and contacts.

Phase 6 — Materials design map, N3 (2 weeks)

- Dense parametric sweeps of k (2–150), D_a (via mobility scaling), C_a and τ_long, using the reference model for density and COMSOL for spot checks.
- Iso-ΔT contours, with Si, Ge, SiGe and GeSn placed on the map.

Phase 7 — Optimisation and inverse design, N1 (2–3 weeks)

- COMSOL Optimization Module. Objective: maximise $T_\mathrm{room} - T_\mathrm{end}$ over $(V_\mathrm{F}, w_\mathrm{rz}, \tau_\mathrm{short}, W_i)$, with constraints $P_\mathrm{el} \le P_\mathrm{max}$ and $V_\mathrm{j} \le V_\mathrm{bi} - 0.1\,\mathrm{V}$. Use a derivative-free method (BOBYQA or Nelder–Mead) for these few parameters.
- Then free-form $\tau(x)$: an interpolation control function, optimised with a gradient method (SNOPT or MMA).
- Expected outcome: an optimal profile that is sharply concentrated beyond x^*, as Eq. (7) predicts.

Verification at every phase

- Energy closure η_E.
- Mesh and tolerance convergence.
- Reproduction of the analytic limits: the length independence of Eq. (9) at low bias, the $1/W_i$ dependence of Eq. (13) at the Auger limit, and the uniform-lifetime heating theorem.
- Agreement with the reference model within the established factor.
- Sign reversal between devices with and without the zone.

7 Conclusions

1. The heat a PIN junction absorbs per injected pair is $\Pi_\mathrm{tot} = 3k_\mathrm{B}T + 2k_\mathrm{B}T\ln(\sqrt{N_c N_v}/n)$. It does not depend on the band gap. Only about the innermost 14% (Si) to 21% (Ge) of each half of the device is a place where recombination produces net cooling. Uniform recombination always heats.
2. The temperature drop is bounded by $n_\mathrm{b} D_\mathrm{a} \Pi_\mathrm{side}/k$, and Auger recombination caps n_b. This gives $\Delta T_\mathrm{max} \propto (D_\mathrm{a}\Pi_\mathrm{side})^{3/2}/(kW_i\sqrt{E_\mathrm{pair} C_\mathrm{a}})$. For silicon this is 10 mK–40 mK at the optimum bias; the thermoelectric view gives $ZT \approx 0.002$.
3. Germanium improves D_a/k by 8.9× and the Auger-limited figure of merit by 13–27×, and it operates at about half the voltage with a higher COP. The reference model predicts 0.15 K–0.40 K, an order of magnitude more than silicon but still below a kelvin.
4. Kelvin-scale bipolar cooling requires $D_\mathrm{a}/k \gtrsim 8 \times 10^{-4}\,\mathrm{m^3\,K/J}$: germanium-like carriers in a lattice with $k \lesssim 5\,\mathrm{W/(m\,K)} - -10\,\mathrm{W/(m\,K)}$ and long lifetimes. GeSn alloys are the most promising candidate, and they are the basis of the most ambitious proposal (N3).

Data and code. All COMSOL drivers, the reference model, the post-processing and the parameter files are in the project repository (`config/`, `comsol/python/`, `reference/`, `postprocess/`). The silicon and germanium numbers of Section 4 and of proposal N3 are reproduced by `python reference/si_vs_ge.py`, which runs `reference/reference_model.py` with the material parameters of Table 2.

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